European Electricity Dependency Map

ENTSO-E 2025 - dependency mapping

22 August 2026 | Supervisor: Stefan

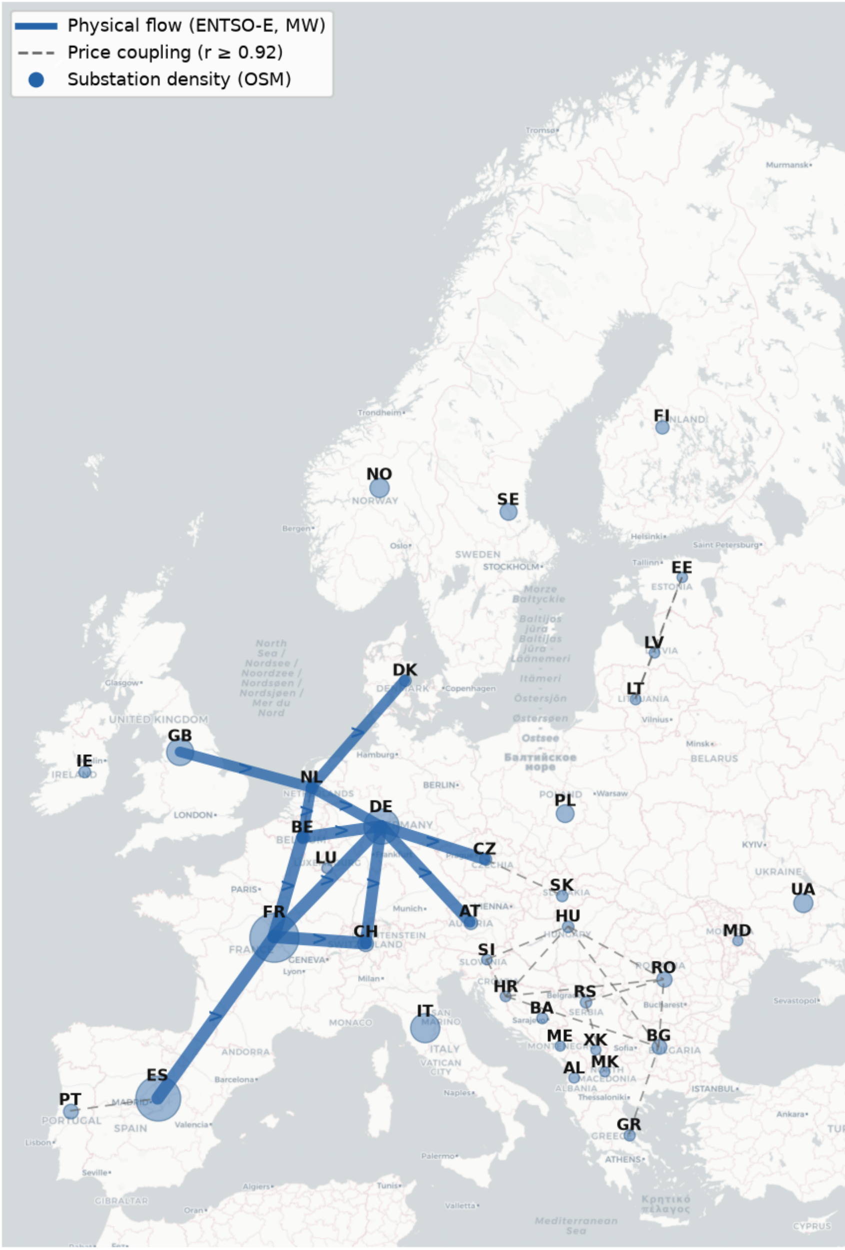
I collected ENTSO-E 2025 time-series for European bidding zones, validated cross-border flows, built a dependency graph, and added PyPSA-Eur grid and GEM plant data as reference layers. Dynamic edges come from ENTSO-E; static data is geographic context only.

Main results

- Strongest price coupling: LT-LV (r~0.998), BG-RO, HR-SI, ES-PT (r~0.978).
- Graph: 45 zones, 68 price edges (r>=0.85), 89 flow edges, 177 structural edges.
- 171/177 directed flow pairs are duplicate aggregate exports in ENTSO-E; only PT-ES and IE_SEM-GB are clearly distinct.
- Static: 9162 OSM lines (PyPSA-Eur), 55,720 GEM units (August 2026 GIPT).
- GB prices missing for 2025 with current codes. German prices use zone DE_LU.

1. Map - flows and price coupling

Blue lines: top cross-border flows (canonical edges, mean |MW|). Grey dashed: price coupling (r>=0.92). Circles: OSM substation count. Basemap: Carto Positron.



2. Bidding zones

Zone codes in all parquet files. Use DE_LU for Germany.

Zone

AL	Albania	AL	No
AT	Austria	AT	No
BA	Bosnia and Herzegovina	BA	No
BE	Belgium	BE	No
BG	Bulgaria	BG	No
CH	Switzerland	CH	No
CZ	Czech Republic	CZ	No
DE_LU	Germany/Luxembourg	DE	Yes
DK_1	Denmark West (DK1)	DK	Yes
DK_2	Denmark East (DK2)	DK	Yes
EE	Estonia	EE	No
ES	Spain	ES	No
FI	Finland	FI	No
FR	France	FR	No
GB	Great Britain	GB	No
GR	Greece	GR	No
HR	Croatia	HR	No
HU	Hungary	HU	No
IE_SEM	Ireland (SEM)	IE	Yes
IT_CNOR	Italy Centre-North	IT	Yes
IT_CSUD	Italy Centre-South	IT	Yes
IT_NORTH	Italy North	IT	Yes
IT_SARD	Italy Sardinia	IT	Yes
IT_SICI	Italy Sicily	IT	Yes
IT_SUD	Italy South	IT	Yes
LT	Lithuania	LT	No
LV	Latvia	LV	No
ME	Montenegro	ME	No
MK	North Macedonia	MK	No
NL	Netherlands	NL	No
NO_1	Norway NO1	NO	Yes
NO_2	Norway NO2	NO	Yes
NO_3	Norway NO3	NO	Yes
NO_4	Norway NO4	NO	Yes
NO_5	Norway NO5	NO	Yes
PL	Poland	PL	No
PT	Portugal	PT	No
RO	Romania	RO	No
RS	Serbia	RS	No
SE_1	Sweden SE1	SE	Yes
SE_2	Sweden SE2	SE	Yes
SE_3	Sweden SE3	SE	Yes
SE_4	Sweden SE4	SE	Yes
SI	Slovenia	SI	No
SK	Slovakia	SK	No

3. ENTSO-E datasets

File	Type	Unit	Description	Coverage
day_ahead_prices	A44 / 12.1.D	EUR/MWh	Hourly market clearing price for next-day delivery	43 zones
actual_load	A65 / 6.1.A	MW	Total electricity demand in the zone	44 zones
generation_actual	A75 / 16.1.B&C	MW	Hourly output split by fuel type (PSR)	41 zones
imbalance_prices	A85 / 17.1.G	EUR/MWh	Balancing energy price (system stress)	40 zones
crossborder_flows	A11 / 12.1.G	MW	Physical power flow between two zones	171 pairs
scheduled_exchanges	A09 / 12.1.E	MW	Commercially scheduled cross-border trade	171 pairs
net_transfer_capacity	A61 / 11.1	MW	Day-ahead max transferable capacity (NTC)	Partial

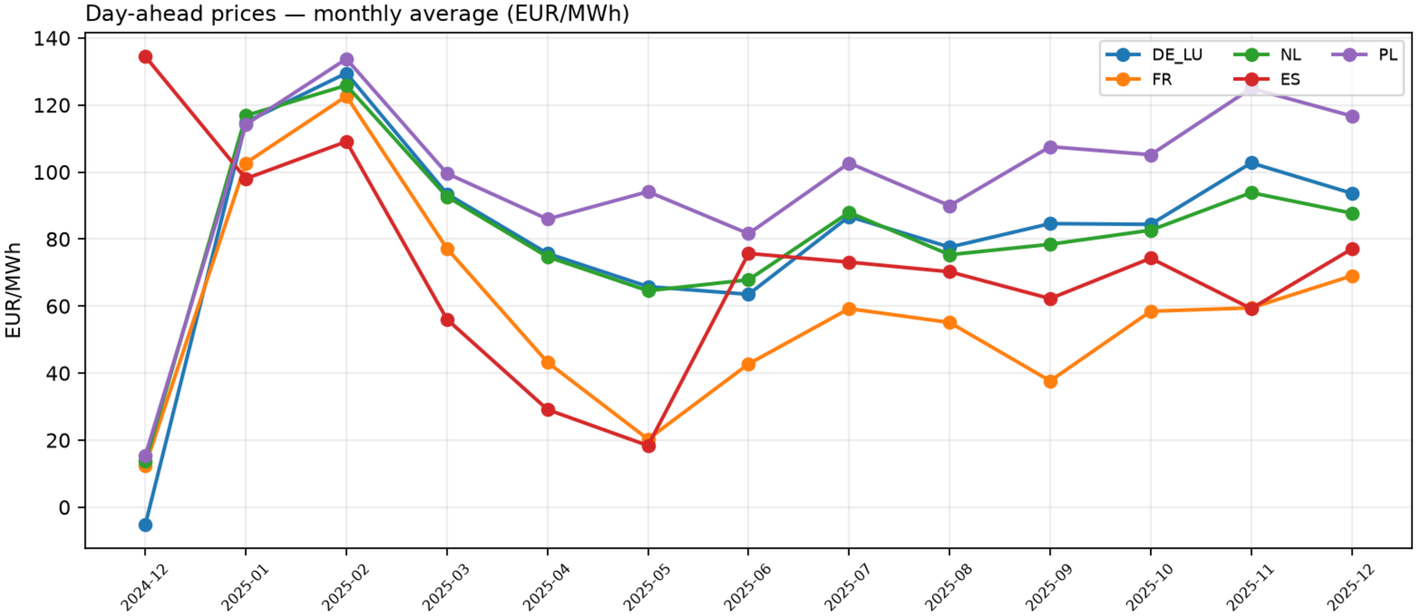
4. Columns and fuel types

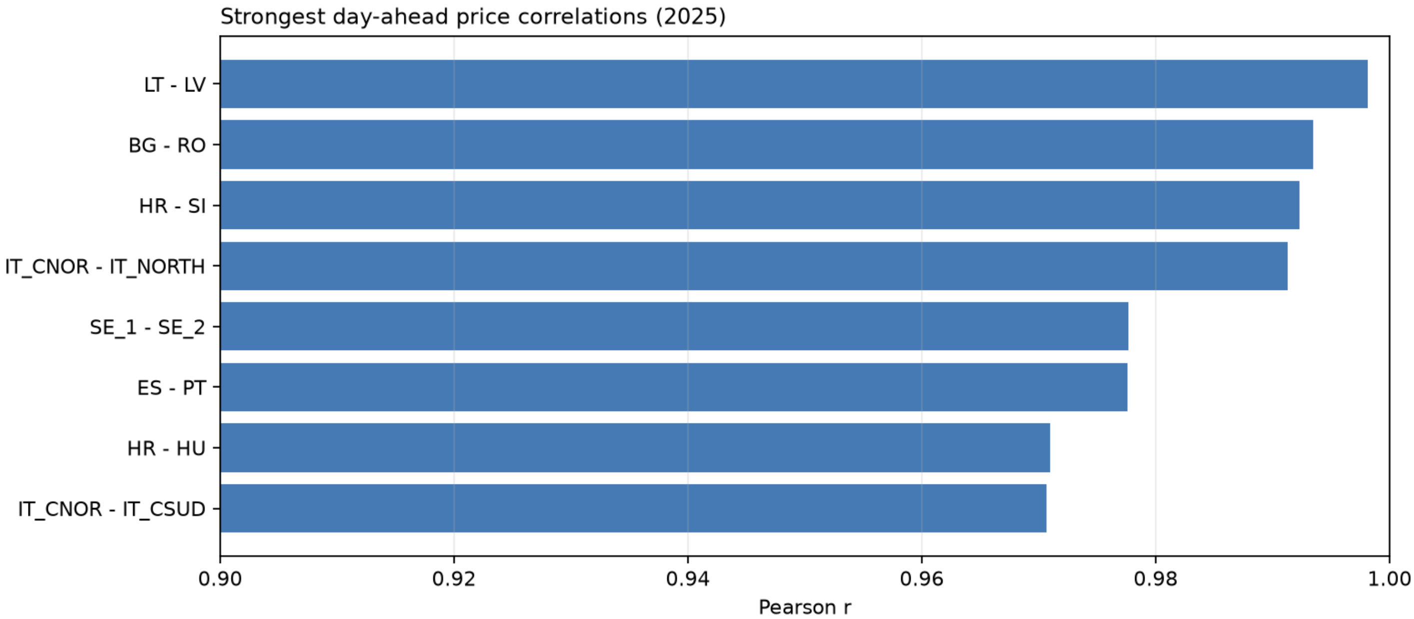
Column	Applies to	Type	Meaning
timestamp	All	UTC datetime	Start of measurement interval
value	All	float	Main numeric value (see unit per dataset)
zone	Zone-level	string	Bidding zone code (e.g. DE_LU, FR, IT_NORTH)
from_zone / to_zone	Border pairs	string	Directed flow: power exported from -> imported to
psr_type	Generation	string	Fuel type: Nuclear, Solar, Wind Onshore, Fossil Gas, ...
currency	Prices	string	EUR
price_unit	Prices	string	MWH (= EUR per MWh)
quantity_unit	Load/flows	string	MW (megawatts)

PSR	Name	Category
B14	Nuclear	Low-carbon baseload
B16	Solar	Renewable
B19	Wind Onshore	Renewable
B18	Wind Offshore	Renewable
B04	Fossil Gas	Fossil
B05	Fossil Hard coal	Fossil
B02	Fossil Brown coal/Lignite	Fossil
B11	Hydro Run-of-river	Renewable
B12	Hydro Water Reservoir	Renewable
B10	Hydro Pumped Storage	Storage
B01	Biomass	Renewable
B09	Geothermal	Renewable

5. Price analysis (2025)

High day-ahead price correlation indicates market integration. I use r>=0.85 as the price-coupling threshold in the graph.





6. Flow validation

Duplicate outgoing series (>=95% identical from the same zone) affect most TSOs.

Status

duplicate_export	171	Same as another outgoing border
partial	2	Coverage < 90%
missing	4	No data
ok	0	Distinct series
graph edges	89	One edge per border

Top flow edges (mean |MW|)

Border	MW	Type	Check
FR->BE	1,598	physical flow canonical	duplicate_export
FR->CH	1,598	physical flow canonical	duplicate_export
FR->DE_LU	1,598	physical flow canonical	duplicate_export
FR->ES	1,598	physical flow canonical	duplicate_export

7. Dependency graph

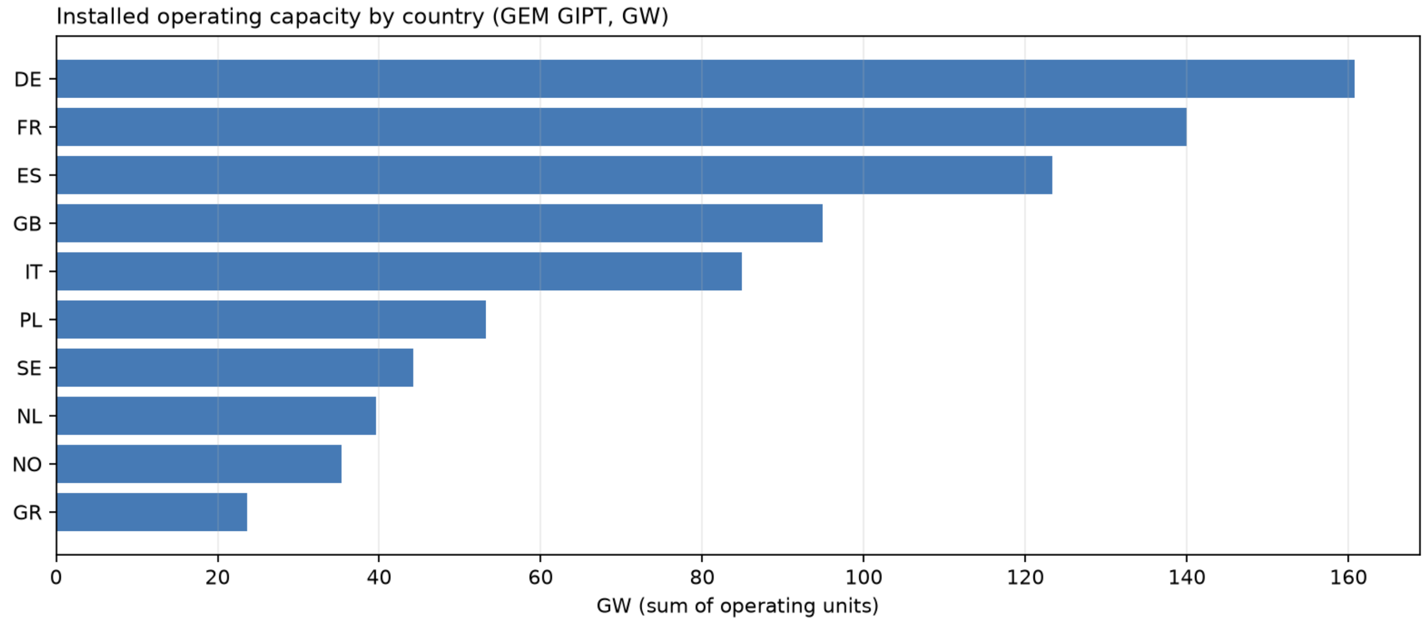
Layer	Count	Source
Price coupling	68	prices_2025 (r>=0.85)
Physical flow	89	flows_2025, canonical
Structural	177	zones.yaml
Nodes	45	zones + OSM centroids

8. Static infrastructure

PyPSA-Eur OSM grid and GEM GIPT (Aug 2026, CC BY 4.0) - reference layers only, not in edge weights.

Source

PyPSA-Eur OSM	9162 lines	Substation density
GEM GIPT	55,720 units	Country capacity on nodes



Operating GEM capacity by country (GW).